

BANGALORE INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MCA

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IntelliExam AI

**Intelligent Assessment, Proctoring and Placement
Preparation System**

Hemanth Kumar K (MCA)

PROJECT GUIDES
Madhu H K

Introduction

Traditional exam engines only test and score. IntelliExam AI transforms testing into a closed-loop diagnostic that drives targeted learning and direct career placement.

1. Online Examinations

Reliable web and LAN testing with low latency.

2. AI-Based Evaluation

Objective rubrics for complex descriptive answers.

3. Smart Proctoring

Vision-backed multi-face and device anomaly alerts.

4. Personalized Learning

Adaptive modules targeting diagnosed weak topics.

5. Resume Intelligence

ATS compatibility scoring and skill-gap matching.

6. AI Mock Interviews

Practise and get trained for real world jobs

Why the Name — "IntelliExam AI"

Intelli

Represents intelligence and automation, ensuring smart and efficient processes.

Exam

Reflects our core focus on assessment and comprehensive examination management.

AI

Highlights the pivotal role of artificial intelligence in evaluation, recommendations, proctoring, and career guidance.

Core Project Objectives

1. Automate the evaluation of descriptive-answer questions, providing faster, objective feedback.
2. Enhance examination security through advanced AI-based proctoring, ensuring academic integrity.
3. Accurately identify student learning gaps to target areas needing improvement.
4. Analyze and optimize student resumes to boost their career readiness and marketability.
5. Provide essential practice for typing, voice, and coding interviews to prepare students for real-world scenarios.
6. Generate comprehensive performance insights for both students and admins.

Existing System

Current educational software stacks suffer from fragmented operations and high faculty workload.

1. Fragmented Workflows

Separate tools for exams, learning portals, and campus placements with zero data sync.

3. Weak Proctoring

Basic or no proctoring; limited cheating-detection capabilities.

2. Manual Evaluation

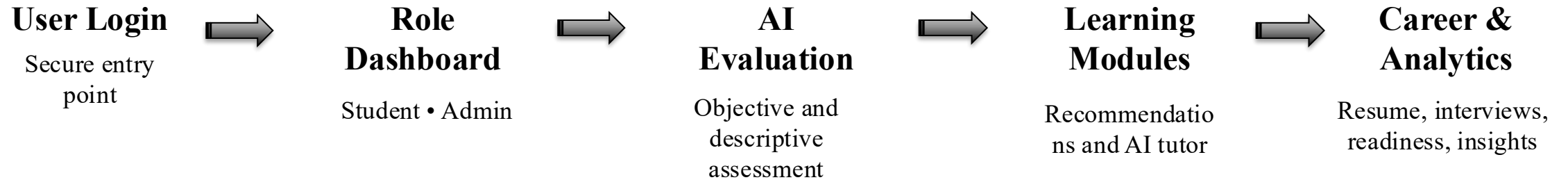
High human workload and delayed feedback for descriptive answers.

4. Insufficient Insights

Scorecards display percentages without mapping root skill deficiencies or hiring market criteria.

Proposed System — IntelliExam AI

IntelliExam AI connects the major stages of the student journey through a shared platform and role-based workflows.



IntelliExam AI connects examination, learning and career preparation through a single AI-powered platform.

User Login: Provides secure access to the platform.

Role Dashboard: Displays features according to the user role—Student, Faculty, Admin

AI Evaluation: Evaluates objective and descriptive answers and provides feedback.

Learning Modules: Identifies learning gaps and recommends resources, roadmaps and AI-tutor support.

Career & Analytics: Supports resume analysis, mock interviews, placement readiness and institutional insights.

Literature Survey

1 AI-Based Proctoring

Face/behavior detection, multi-face and mobile-phone detection approaches.

Paper :- “Enhancing Academic Integrity in Online Assessments: Introducing an Effective Online Exam Proctoring Model Using YOLO”

Authors: Tripty Singh, Rekha R. Nair, Tina Babu and Prakash Duraisamy

Published in: *Procedia Computer Science*, 2024

Paper :- “iExam: A Novel Online Exam Monitoring and Analysis System Based on Face Detection and Recognition”

Authors: Xu Yang, Daoyuan Wu, Xiao Yi, Jimmy H. M. Lee and Tan Lee

Published: 2022

2 Automated Answer Evaluation

NLP techniques for relevance, and automated evaluation.

Paper :-“Automatic Assessment of Text-Based Responses in Post-Secondary Education: A Systematic Review”

Authors: Rujun Gao, Hillary E. Merzdorf, Saira Anwar, M. Cynthia Hipwell and Arun R. Srinivasa

Published in: *Computers and Education: Artificial Intelligence*, 2024

Paper :- “Automated Grading and Feedback Tools for Programming Education: A Systematic Review”

Authors: Marcus Messer, Neil C. C. Brown, Michael Kölling and Miaoqing Shi

Published: 2023

3 **Personalized Learning**

Knowledge tracing, adaptive recommendation, and roadmap generation.

Paper :- “Enhanced Personalized Learning Exercise Question Recommendation Model Based on Knowledge Tracing”

Authors: Pei Pei, Rodolfo C. Raga Jr. and Mideth Abisado

Published in: *International Journal of Advances in Intelligent Informatics*, 2024

Paper :- “Knowledge Modeling via Contextualized Representations for LSTM-Based Personalized Exercise Recommendation”

Authors: Feng Zhang, Xuguang Feng and Yibing Wang

Published in: *Information Sciences*, 2020

4 **Resume & Career Analytics**

Parsing, ATS-matching, and skill-gap analysis using NLP.

Paper :- “NLP-Based Bi-Directional Recommendation System: Towards Recommending Jobs to Job Seekers and Resumes to Recruiters”

Authors: Suleiman Ali Alsaif, Minyar Sassi Hidri, Imen Ferjani, Hassan Ahmed Eleraky and Adel Hidri

Published in: *Big Data and Cognitive Computing*, 2022

Paper :- “NLP-Based Resume Screening and Job Recruitment Portal”

Authors: Rohit Kadam, Gloria Suhas, Uroosa Mukri and Suraj Khandare

Published in: *Data Intelligence and Cognitive Informatics*, Springer, 2022

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Authentication, session security, data isolation, and compliance considerations.

Paper 09 :- “Privacy in Online Proctoring Systems in Higher Education: Stakeholders’ Perceptions, Awareness and Responsibility”

Authors: Chantal Mutimukwe, Olga Viberg, Cormac McGrath and Teresa Cerratto-Pargman

Published in: *Journal of Computing in Higher Education*, 2026

Paper 10:- “Examining the Examiners: Students’ Privacy and Security Perceptions of Online Proctoring Services”

Authors: David G. Balash, Dongkun Kim, Darikia Shaibekova, Rahel A. Fainchtein,
Micah Sherr and Adam J. Aviv

Published in: *SOUPS 2021*

Research to Implementation

AI-Based Proctoring (Implementation)

- 1 Face detection / absence detection / multi-face alerts
- 2 Mobile-phone and object-in-frame detection
- 3 Application-switch and attention drift monitoring
- 4 Suspicious-activity scoring with automated critical-violation submission

Automated Answer Evaluation (Implementation)

- 1 NLP scoring: correctness, relevance, and completeness
- 2 Technical correctness checks for domain-specific answers
- 3 Gibberish and plagiarism detection
- 4 Structured rubric outputs and improvement suggestions

Resume & Career Intelligence (Implementation)

1. Resume parsing from PDF, DOCX and TXT files
2. Technical skill extraction and role-specific skill classification
3. ATS-style resume scoring and job-role matching
4. Resume optimization with skill-gap analysis and personalized career roadmaps

Security & Privacy (Implementation)

1. JWT-based authentication and secure user login
2. Role-Based Access Control for Student, Faculty, Admin and Placement Team
3. Student-wise data isolation and protected examination records
4. Secure API routes, session management and input validation

Technology Stack

The platform combines a conventional web application stack with AI services, computer vision, and a desktop client for secure examination workflows.

01	Frontend	HTML5, CSS3, JavaScript — responsive exam and dashboard UI.
02	Backend	Node.js + Express.js for APIs, authentication, and real-time exam flows.
03	Database	MySQL for structured records: users, exams, results, and audit logs.
04	AI Services	Google Gemini API and OpenAI-compatible services for NLP, scoring, and conversational tutors.
05	Proctoring & Security	Computer vision (face/object detection), JWT + RBAC authentication, secure tunnels for desktop clients.
06	Desktop App & Networking	Electron desktop client with optional LAN access and Cloudflare secure tunnel for remote proctored exams.



Thank You

Hemanth Kumar K (MCA)

Bangalore Institute of Technology

Madhu H K